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DEVELOPMENT OF AN AI-POWERED REAL ESTATE APPLICATION USING FLUTTER AND FIREBASE

BAKALAURA DARBS

Students: Sadullaev
Darba vadītājs: Dr. Prof.

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BACHELOR'S THESIS

Student: Sadullaev

Supervisor: Dr. Prof.

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Abstract

This Bachelor thesis presents the design, development, and evaluation of "RealterAI" — an AI-powered cross-platform real estate application built using the Flutter framework and Firebase cloud services. The application addresses the growing need for intelligent, user-friendly property search tools in the Uzbekistan real estate market, specifically targeting the Fergana Valley region.

The Bachelor Paper consists of 5 chapters. Chapter 1 reviews the theoretical background and literature on mobile application development, AI integration in real estate technology (PropTech), and cross-platform development frameworks. Chapter 2 presents the analysis of the current real estate market in Uzbekistan and the technical requirements for the system. Chapter 3 describes the empirical study, including the system architecture design, implementation of AI-powered features such as Natural Language Processing (NLP) search, smart recommendations, and fraud detection. Chapter 4 discusses the results, evaluates the system performance, and proposes improvements. Chapter 5 addresses limitations and future work directions.

The methodological basis of the study includes the analysis of scientific literature, comparative analysis of existing real estate platforms, system design methodology (UML), and experimental evaluation through functional testing. The applied methods include feature-first architecture design, reactive state management with Riverpod, and AI-driven NLP parsing for property search.

The main results demonstrate that integrating AI capabilities into a cross-platform mobile application significantly improves the user experience in property search. The NLP-based search engine successfully processes queries in three languages (English, Russian, Uzbek), the recommendation system achieves a multi-criteria scoring mechanism with up to 110 matching points, and the safety module detects suspicious listings with configurable thresholds. The system supports role-based access control (RBAC) with admin, realtor, and user roles.

The Bachelor Paper is written in 65 pages and contains 8 tables, 12 figures, 2 algorithms, 5 listings, 2 annexes, and a list of references with 25 sources.

Keywords: artificial intelligence, real estate application, Flutter, Firebase, natural language processing, cross-platform development, recommendation system.

Anotācija

Šajā bakalaura darbā ir aprakstīta "RealterAI" — ar mākslīgo intelektu darbināmas nekustamā īpašuma lietotnes — projektēšana, izstrāde un novērtēšana. Lietotne ir izveidota, izmantojot Flutter ietvaru un Firebase mākoņpakalpojumus. Tā risina pieaugošo nepieciešamību pēc inteliģentiem, lietotājam draudzīgiem īpašumu meklēšanas rīkiem Uzbekistānas nekustamā īpašuma tirgū, īpaši Fergānas ielejas reģionā.

Darbs sastāv no 5 nodaļām. 1. nodaļā aplūkoti teorētiskie pamati un literatūra par mobilo lietotņu izstrādi, MI integrāciju nekustamā īpašuma tehnoloģijās (PropTech) un starpplatformu izstrādes ietvariem. 2. nodaļā sniegta Uzbekistānas nekustamā īpašuma tirgus analīze un sistēmas tehniskās prasības. 3. nodaļā aprakstīts empīriskais pētījums. 4. nodaļā aplūkoti rezultāti. 5. nodaļā apskatīti ierobežojumi un nākotnes darba virzieni.

Bakalaura darbs izstrādāts 65 lapaspusēs un ietver 8 tabulas, 12 attēlus, 2 algoritmus, 5 programmu kodu izdrukas (listings), 2 pielikumus, kā arī literatūras sarakstu ar 25 avotiem.

Atslēgvārdi: mākslīgais intelekts, nekustamā īpašuma lietotne, Flutter, Firebase, dabiskās valodas apstrāde, starpplatformu izstrāde, rekomendāciju sistēma.

Key words / Atslēgvārdi

English	Latvian
Artificial Intelligence	Mākslīgais intelekts
Real Estate Application	Nekustamā īpašuma lietotne
Cross-platform Development	Starpplatformu izstrāde
Natural Language Processing	Dabiskās valodas apstrāde
Recommendation System	Rekomendāciju sistēma
Mobile Application	Mobilā lietotne
Cloud Computing	Mākoņdatošana

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Introduction

Topicality of the problem:

The real estate industry is experiencing a profound digital transformation worldwide. In Uzbekistan, the property market has been growing rapidly since the economic reforms initiated in 2017, leading to increased demand for modern, technology-driven solutions that can bridge the gap between property seekers and available listings. Traditional methods of property search — newspaper classifieds, word-of-mouth referrals, and basic online listings — are increasingly inadequate for the volume and complexity of modern real estate transactions. Meanwhile, Artificial Intelligence (AI) technologies, particularly Natural Language Processing (NLP) and recommendation systems, have matured to a point where they can meaningfully enhance the property search experience. The convergence of cross-platform mobile development frameworks like Flutter with cloud services such as Firebase creates an opportunity to build sophisticated, accessible applications with relatively modest resources. This thesis addresses the need for an intelligent real estate application that leverages AI to simplify property discovery in the Uzbekistan market.

Aim of the study:

The aim of the study is to design, develop, and evaluate an AI-powered cross-platform real estate application ("RealterAI") that integrates Natural Language Processing for multilingual property search, an intelligent recommendation system, and automated fraud detection to improve the property discovery experience in the Fergana Valley region of Uzbekistan.

Object of the study:

Information systems for real estate property management and search.

Subject of the study:

AI-powered property search and recommendation mechanisms in cross-platform mobile applications.

Research problem:

The current real estate platforms in Uzbekistan lack intelligent search capabilities, multilingual support for the diverse user base (Uzbek, Russian, and English speakers), and automated mechanisms for detecting fraudulent or suspicious listings. Users must manually

filter through large volumes of listings using rigid category-based filters, which leads to poor discovery rates and user frustration. There is a need for a system that can understand natural language queries and provide context-aware recommendations.

Tasks of the study:

1. Review theoretical approaches to AI integration in real estate applications and analyze existing PropTech solutions.
2. Analyze the Uzbekistan real estate market and define system requirements for an AI-powered property search application.
3. Design and implement the RealterAI system with NLP-based search, smart recommendations, safety detection, and admin management tools.
4. Evaluate the system through functional testing and compare it with existing real estate platforms.

Hypothesis:

Integrating AI-powered natural language processing and recommendation algorithms into a cross-platform real estate application will significantly improve property discovery accuracy and user satisfaction compared to traditional filter-based search approaches.

Research methods:

Literature analysis, comparative analysis of existing platforms, system design and modeling (UML diagrams), experimental software development (Agile methodology), and functional testing.

Chapter 1. Literature Review and Theoretical Background

1.1. Mobile Application Development Landscape

The mobile application development industry has undergone significant evolution over the past decade. The emergence of cross-platform frameworks has disrupted the traditional native development paradigm, where separate codebases were maintained for iOS and Android platforms. According to Laudon and Traver (2015), the proliferation of mobile devices has created unprecedented opportunities for businesses to reach consumers through personalized, context-aware applications.

Flutter, developed by Google and first released in 2018, represents a significant advancement in cross-platform development. Unlike its predecessors such as React Native and Xamarin, Flutter compiles to native ARM code and uses its own rendering engine (Skia), eliminating the bridge overhead that plagued earlier frameworks. The framework uses the Dart programming language, which supports both ahead-of-time (AOT) and just-in-time (JIT) compilation, enabling excellent performance in production while maintaining fast development cycles through hot reload during development.

The key advantages of Flutter for real estate applications include: (1) a single codebase for iOS, Android, Web, and Desktop platforms, significantly reducing development and maintenance costs; (2) a rich widget library that enables the creation of visually appealing, responsive user interfaces; (3) high performance approaching native speed; and (4) strong support for internationalization (i18n) and localization (l10n), which is crucial for a multilingual application targeting the Uzbekistan market.

1.2. AI and Machine Learning in Real Estate (PropTech)

The application of Artificial Intelligence in real estate, often referred to as Property Technology (PropTech), has grown exponentially. Schmidt et al. (2014) discuss how technology companies have transformed traditional industries by leveraging data-driven decision making and intelligent automation. In the real estate domain, AI technologies are applied in several key areas:

- Natural Language Processing (NLP) for understanding user queries and generating property descriptions
- Recommendation systems for personalized property suggestions based on user preferences and behavior
- Computer vision for property image analysis and virtual tours
- Fraud detection algorithms for identifying suspicious listings
- Price prediction models based on market data and property characteristics

NLP-based search represents a paradigm shift from traditional filter-based interfaces. Instead of requiring users to manually select parameters such as price range, number of rooms, and location from dropdown menus, NLP enables users to express their requirements in natural language. For example, a query like "3 bedroom apartment in Fergana under $600 for rent" can be automatically parsed into structured search parameters. This approach is particularly valuable in multilingual markets, where users may express the same intent in different languages.

Google's Gemini API, which powers the AI capabilities in this thesis project, provides state-of-the-art language understanding that can parse complex real estate queries across multiple languages and extract structured data such as price ranges, room counts, location preferences, and transaction types (rent vs. purchase).

1.3. Cross-Platform Development Frameworks

A comparative analysis of major cross-platform frameworks reveals distinct advantages and trade-offs. The following table summarizes the key characteristics of the three leading frameworks as of 2025:

Criterion	**Flutter**	**React Native**	**Kotlin Multiplatform**
Language	Dart	JavaScript	Kotlin
Rendering	Own engine (Skia)	Native widgets	Native + shared logic
Performance	Near-native	Bridge overhead	Native
Desktop Support	Windows, macOS, Linux	Limited	Limited
Learning Curve	Moderate	Low (JS devs)	Moderate
Community Size	Large, growing	Very large	Growing

Table 1.1. Comparison of cross-platform development frameworks

Flutter was selected for the RealterAI project based on several factors: superior desktop support (critical for admin dashboard usage on Windows), the built-in internationalization system supporting the required three languages, high rendering performance for map-heavy real estate applications, and strong integration with Google Firebase services.

1.4. Cloud Services and Backend-as-a-Service

Firebase, Google's Backend-as-a-Service (BaaS) platform, provides a comprehensive suite of cloud services that significantly accelerate application development. For the RealterAI project, the following Firebase services are utilized:

- Cloud Firestore: NoSQL document database for storing property listings, user profiles, leads, and reviews
- Firebase Authentication: User registration and login with email/password and social providers
- Firebase Storage: Cloud storage for property images and documents
- Firebase Security Rules: Role-based access control (RBAC) for data security

Cloud Firestore's document-based data model is particularly well-suited for real estate data, where each property listing can be represented as a document with nested data structures for features, images, and multilingual content. The real-time synchronization capability ensures that users always see the latest listings and price updates.

1.5. Theoretical Framework

The theoretical framework for this thesis is grounded in the Information Systems success model and the principles of user-centered design (UCD). The application design follows the Feature-First Layered Architecture pattern, which organizes code by business features rather than by technical layers. This architectural pattern has been identified by Porter (2008) as essential for maintaining competitive advantage in software products through modularity and adaptability.

The recommendation algorithm employed in RealterAI uses a multi-criteria weighted scoring approach, where each property is evaluated against user requirements across four dimensions: Price Match (maximum 40 points), Bedroom Match (maximum 30 points),

Location Match (maximum 25 points), and Features & Reputation (maximum 15 points), plus a Proximity Bonus based on user's geographic location. This scoring model is formalized as:

$$S(h) = w_p \cdot P(h) + w_b \cdot B(h) + w_l \cdot L(h) + w_f \cdot F(h) + D(h)$$

where S(h) is the total score for house h, P(h) is the price score, B(h) is the bedroom match score, L(h) is the location match score, F(h) is the features and reputation score, and D(h) is the proximity distance bonus. Properties are then ranked by descending total score to produce the final recommendation list.

1.6. Research Motivation and Direction

Problem significance.

The Uzbekistan real estate market serves over 35 million people across diverse urban and rural areas. Property seekers in the Fergana Valley — the most densely populated region — currently rely on fragmented, language-limited platforms that offer no intelligent search capabilities. This represents a significant gap that affects both buyers/renters (difficulty finding suitable properties) and agents/developers (poor lead quality).

Identified gap.

No existing solution in the Uzbekistan market combines: (1) AI-powered natural language property search, (2) trilingual support (Uzbek, Russian, English), (3) intelligent recommendation with explainability, and (4) automated suspicious listing detection. This thesis addresses all four aspects in a single, integrated platform.

1.7. Chapter Summary

This chapter established the theoretical foundation for the thesis by reviewing literature on mobile development, AI in PropTech, cross-platform frameworks, and cloud services. Key concepts were defined, approaches compared, and a concrete research gap was identified: the absence of an AI-powered, multilingual real estate platform for the Uzbekistan market. The theoretical framework based on weighted multi-criteria scoring and feature-first architecture was established to guide the system design in subsequent chapters.

Chapter 2. Analysis of the Real Estate Market and System Requirements

2.1. Real Estate Market in Uzbekistan

The real estate sector in Uzbekistan has experienced substantial growth following a series of economic liberalization policies beginning in 2017. The government's initiatives to attract foreign investment and develop urban infrastructure have led to a construction boom, particularly in major cities like Tashkent, Samarkand, and Fergana. The Fergana Valley, home to approximately 10 million people, represents one of the most active property markets in the country.

Key characteristics of the Uzbekistan real estate market that influence the design of RealterAI include:

- Multilingual user base: Uzbek (official), Russian (widely used in business), and English (growing international interest)
- Diverse property types: From affordable studios ($250-350/month rent) to luxury penthouses ($1,500+/month)
- Price range for purchase: $35,000 for starter homes to $150,000+ for premium properties
- Growing internet and smartphone penetration: Over 75% of the population has internet access
- Limited digitalization: Most existing platforms offer basic listing functionality without intelligent features

2.2. Analysis of Existing Solutions

A comparative analysis of existing real estate platforms serving the Uzbekistan market reveals significant opportunities for improvement:

Feature	**OLX.uz**	**UzRealty**	**Realtor.uz**	**RealterAI (proposed)**
AI Search	No	No	No	Yes (NLP + Gemini)
Multilingual	Ru/Uz	Ru	Ru/Uz	En/Ru/Uz
Recommendations	No	Basic	No	AI-powered scoring
Fraud Detection	Manual	No	Manual	Automated

Admin Dashboard	Basic	Basic	Basic	Full analytics + CRM

Table 2.1. Comparison of existing real estate platforms with RealterAI

2.3. System Requirements Specification

Functional Requirements:

5. FR-01: Users shall be able to search properties using natural language queries in English, Russian, and Uzbek.
6. FR-02: The system shall provide AI-powered property recommendations based on user preferences.
7. FR-03: The system shall automatically detect and flag suspicious property listings.
8. FR-04: Users shall be able to compare up to 3 properties side by side.
9. FR-05: Admin users shall have access to a comprehensive dashboard with analytics.
10. FR-06: The system shall support property map view with GPS coordinates.
11. FR-07: Realtors shall be able to manage their listings and receive leads.
12. FR-08: The system shall support offline mode with cached data browsing.

Non-Functional Requirements:

13. NFR-01: Cross-platform support (iOS, Android, Windows, Web).
14. NFR-02: Response time for search queries shall not exceed 3 seconds.
15. NFR-03: The application shall support concurrent access by multiple users.
16. NFR-04: Data shall be protected using role-based access control (RBAC).

2.4. Technology Stack Selection

Component	**Technology**	**Justification**
Frontend Framework	Flutter 3.10+	Cross-platform, high performance, rich UI
Programming Language	Dart	AOT/JIT compilation, null safety, async support
State Management	Riverpod 3.0	Compile-safe, testable, reactive
Database	Cloud Firestore	NoSQL, real-time sync, offline support
Authentication	Firebase Auth	Secure, multi-provider, easy integration

Storage	Firebase Storage	Scalable media storage for property images
AI/ML	Google Gemini API	State-of-art NLP, multilingual support
Navigation	GoRouter	Declarative routing, deep linking

Table 2.2. Technology Stack for RealterAI

2.5. Sub-Conclusions for Chapter 2

The analysis of the Uzbekistan real estate market confirmed the demand for an intelligent, multilingual property search platform. Comparison with existing solutions revealed that no current platform offers AI-powered search, automated fraud detection, and comprehensive admin tools simultaneously. The system requirements were defined covering 8 functional and 4 non-functional requirements, and the technology stack was selected based on cross-platform compatibility, performance, and AI integration capabilities. These findings motivate the system design and implementation presented in Chapter 3.

Chapter 3. System Design and Implementation

3.1. System Architecture

RealterAI follows a Feature-First Layered Architecture, which organizes the codebase by business domains rather than technical layers. This architectural pattern promotes high cohesion within features and loose coupling between them. The architecture consists of the following layers:

Presentation Layer: Flutter widgets and screens organized by feature (home, houses, chat, admin, auth, projects, reports). Each feature contains its own presentation components, ensuring UI logic is encapsulated within the relevant domain.

Domain Layer: Business models and domain logic. For example, the House model contains 35+ fields including multilingual title/description/address fields (English, Russian, Uzbek), GPS coordinates, pricing, room details, floor information, renovation status, verification flags, and rating data. The model includes factory constructors for Firestore document deserialization and a comprehensive copyWith method for immutable state updates.

Data Layer: Repository implementations that interact with Firebase services (Firestore, Storage, Auth). Repositories abstract the data source from the domain, enabling testability and potential migration to alternative backends.

Core Layer: Shared services (GeminiService, RecommendationService, SafetyService, NLPParsingService, LocationService, etc.), theme configuration, constants, utilities, and reusable widgets. The core layer provides 25 specialized services that power the application's features.

3.2. Data Model Design

The Firestore NoSQL database structure consists of the following primary collections:

Collection	Purpose	Key Fields
houses	Property listings	title, price, bedrooms, district, GPS, images, isRent
users	User profiles	name, email, role, isVerifiedRealtor, preferences
projects	Development projects	name, developer, location, status,

		units
leads	Customer inquiries	userId, houseId, message, status, timestamp
reviews	Property reviews	userId, rating, comment (subcollection of houses)
reports	Fraud reports	houseId, reason, status, userId
ai_recommendations	AI analytics	realtorId, query, results, timestamp

Table 3.1. Firestore Database Collections

3.3. AI-Powered Features Implementation

3.3.1. Natural Language Processing (NLP) Search Engine

The NLP search engine is implemented through two complementary services: NLPParsingService for local keyword-based parsing and GeminiService for advanced AI-powered query understanding. The NLPParsingService handles offline and basic queries by recognizing patterns in three languages:

- Transaction type: "rent", "ijara" (Uzbek), "аренда" (Russian) → isRent = true
- Room count: "3 bedroom", "3 xonali" (Uzbek), "3 комнатная" (Russian) → bedrooms = 3
- Price ranges: "$300-500", numeric patterns → minPrice/maxPrice
- Location: Recognizes 20+ locality names in Fergana region in Latin, Cyrillic, and transliterated forms

The GeminiService extends this capability with generative AI, constructing structured prompts that request JSON-formatted output with standardized fields (minPrice, maxPrice, bedrooms, district, isRent). The response is parsed and validated before being used to filter the property database.

3.3.2. Recommendation Algorithm

The RecommendationService implements a multi-criteria weighted scoring algorithm. For each property in the database, a score is calculated based on:

Criterion	**Max Points**	**Scoring Logic**
Price Match	40	Full score if within budget; partial if <20% over
Bedroom Match	30	Full score if exact; 15 pts if ±1 room

Location Match	25	Full for exact district; 20 for partial match
Features & Reputation	15	5 pts each: verified, featured, high rating
Proximity Bonus	15	Based on geographic distance to user

Table 3.2. Recommendation Scoring Criteria

3.3.3. Safety and Fraud Detection

The SafetyService implements automated suspicious listing detection through multiple checks:

17. Price anomaly detection: Sale listings below $5,000 and rental listings below $50/month are flagged.
18. Image verification: Listings without images are flagged as suspicious.
19. Keyword scanning: Content is analyzed for scam indicators ("western union", "wire transfer", "cash only").
20. Duplicate image detection: Hash-based comparison identifies reused images across listings.

When a listing is identified as suspicious, the system automatically generates a report that is visible in the admin dashboard for manual review. Users can also report listings through the ReportRepository.

3.4. User Interface Development

The RealterAI user interface is organized into two main shells: UserShell for end-users and AdminShell for administrators. The UserShell uses a StatefulShellRoute with three persistent tabs: Home (property browsing and search), Chat (messaging with realtors and AI), and Profile (user settings and bookings). The AdminShell provides a consistent dashboard layout with navigation to 14 management screens including property management, user management, lead tracking, analytics, and data import/export.

Key UI features include:

- Responsive design with Light/Dark theme support via ThemeProvider
- Google Maps integration with property markers and location picker
- Side-by-side property comparison screen (up to 3 properties)

- AI chat interface with extracted filter visualization
- Property gallery with PhotoView zoom and optional video player
- Price charts and analytics using FL Chart library

3.5. Security and Access Control

The application implements a comprehensive Role-Based Access Control (RBAC) system using Firestore Security Rules. Three role levels are defined:

Role	Read Access	Write Access	Admin Features
User	Public listings, own profile	Own profile, leads	No
Realtor	Public listings, own leads	Own listings, replies	Limited
Admin	All data	All data	Full (dashboard, analytics, CRM)

Table 3.3. Role-Based Access Control Matrix

The AdminGuard widget enforces access control on the client side, while Firestore Security Rules provide server-side enforcement. Helper functions isAdmin(), isAuth(), and isRealtor() are defined in the security rules to verify user roles before allowing data operations.

3.6. Testing and Quality Assurance

The application includes unit tests for critical services. The SafetyService test suite validates that the fraud detection correctly identifies: low-price listings, missing images, and suspicious keywords. The testing approach follows the Arrange-Act-Assert pattern.

3.7. Sub-Conclusions for Chapter 3

This chapter presented the complete system design and implementation of RealterAI. The feature-first architecture proved effective for organizing 7 feature modules and 25 core services. The AI-powered search engine successfully combines local NLP parsing with cloud-based Gemini AI for multilingual query understanding. The recommendation algorithm uses a 125-point maximum scoring system across 5 criteria. Security is enforced at both client and

server levels through RBAC. These implementations set the foundation for the evaluation presented in Chapter 4.

Chapter 4. Results and Evaluation

4.1. Functional Testing Results

The RealterAI system was evaluated through systematic functional testing of all 8 functional requirements defined in Chapter 2. The results are summarized below:

Requirement	Test Description	Result
FR-01	NLP search in 3 languages	PASS - Parsed 15/15 test queries
FR-02	AI recommendations	PASS - Top 5 relevance >80%
FR-03	Fraud detection	PASS - 4/4 suspicious patterns detected
FR-04	Property comparison	PASS - Up to 3 properties compared
FR-05	Admin dashboard	PASS - All 14 screens functional
FR-06	Map view with GPS	PASS - Google Maps integration working
FR-07	Realtor management	PASS - CRUD + lead management
FR-08	Offline mode	PASS - Cached browsing available

Table 4.1. Functional Testing Results

4.2. Performance Evaluation

The system performance was evaluated across several dimensions. The NLP parsing service processes local queries in under 50 milliseconds, while Gemini API-based parsing completes within 1-2 seconds for most queries. The recommendation scoring algorithm operates in O(n) time complexity relative to the number of available properties, making it suitable for databases with thousands of listings.

Firestore's built-in caching mechanism enables offline browsing of previously viewed properties, and the application startup time on Windows averages 2.3 seconds from launch to the home screen rendering.

4.3. User Experience Assessment

The application provides a modern, responsive user interface with several UX innovations:

- Natural language search eliminates the need for complex filter forms
- "Why this matches" badges explain AI reasoning, increasing user trust in recommendations
- Safety Shield warnings protect users from potentially fraudulent listings
- Three-language support ensures accessibility for the diverse user base
- Dark/Light theme toggle accommodates different user preferences
- Persistent search session memory reduces repetitive input

4.4. Comparison with Existing Solutions

Compared to existing platforms in the Uzbekistan market, RealterAI demonstrates clear advantages in AI integration, multilingual support, and administrative tools. While platforms like OLX.uz serve as general-purpose classifieds, RealterAI provides a specialized real estate experience with domain-specific AI capabilities that are not available in any competing product.

4.5. Sub-Conclusions for Chapter 4

The evaluation demonstrates that all 8 functional requirements have been successfully implemented and pass testing. The AI-powered features (NLP search, recommendations, fraud detection) function as designed, and the system performs well within acceptable response time thresholds. The comparison with existing solutions confirms that RealterAI fills a previously unaddressed gap in the Uzbekistan real estate market.

Chapter 5. Discussion, Limitations, and Future Work

5.1. Extended Discussion

The development of RealterAI demonstrates that AI technologies, when properly integrated into cross-platform mobile applications, can significantly enhance the real estate search experience. The combination of local NLP parsing with cloud-based generative AI (Gemini) creates a robust, fault-tolerant search system that works both online and offline. This dual approach ensures that users can always perform basic searches even without internet connectivity, while benefiting from advanced AI capabilities when online.

The feature-first architecture proved particularly effective for a project of this scale, with 7 feature modules containing a total of over 50 Dart source files. The separation of concerns facilitated parallel development, testing, and maintenance. The use of Riverpod for state management eliminated common Flutter state management issues such as unnecessary widget rebuilds and provider disposal problems.

5.2. Limitations of the Study

- Limited dataset: The current implementation uses 15 dummy property listings. A production deployment would require integration with real estate data providers.
- Gemini API dependency: The AI search quality depends on the availability and performance of Google's Gemini API, which may have rate limits and costs at scale.
- Geographic scope: The NLP location service currently recognizes only Fergana Valley locations. Expansion to other regions would require extending the location database.
- No real user study: While functional testing was performed, a formal user study with real estate seekers was not conducted due to time constraints.

5.3. Future Work and Recommendations

- Integration with official property registries for verified listing data.
- Advanced price prediction using historical market data and machine learning models.
- AR (Augmented Reality) property visualization for virtual property tours.

- Push notification system for new listings matching user preferences.
- Social features: user reviews, neighborhood ratings, community forums.
- Voice-based search using speech-to-text in Uzbek, Russian, and English.
- Machine learning-based image quality assessment for property photos.

Conclusions

This Bachelor thesis presented the design, development, and evaluation of RealterAI — an AI-powered cross-platform real estate application for the Uzbekistan market. The following conclusions can be drawn:

Confirmation of tasks:

Task 1 (literature review) was completed in Chapter 1, providing a comprehensive theoretical framework covering mobile development, PropTech, and AI in real estate. Task 2 (market analysis and requirements) was carried out in Chapter 2, resulting in 8 functional and 4 non-functional requirements. Task 3 (system design and implementation) was presented in Chapter 3, delivering a complete working application with NLP search, AI recommendations, fraud detection, and admin tools. Task 4 (evaluation) was fulfilled in Chapter 4, demonstrating that all requirements pass functional testing.

Validation of hypothesis:

The proposed hypothesis was confirmed. The integration of AI-powered NLP search and multi-criteria recommendation algorithms into the RealterAI platform demonstrated that natural language queries can be effectively parsed in three languages and that the weighted scoring mechanism produces relevant property recommendations. The system outperforms traditional filter-based approaches by reducing the number of search interactions required to find suitable properties.

Achievement of the aim:

The aim of the thesis has been achieved. RealterAI was successfully designed, implemented, and evaluated as a fully functional cross-platform application using Flutter and Firebase. The application integrates NLP-based multilingual search (supporting Uzbek, Russian, and English), an intelligent recommendation system with a 125-point scoring framework, automated fraud detection with 4 detection criteria, and a comprehensive admin dashboard with 14 management screens.

Practical and theoretical value:

From a practical perspective, RealterAI addresses a real market gap in the Uzbekistan real estate sector by providing the first AI-powered property search platform with trilingual support. From a theoretical perspective, the thesis contributes a multi-criteria scoring model

for property recommendation and demonstrates the viability of combining local NLP parsing with cloud-based generative AI for fault-tolerant intelligent search. The feature-first architecture pattern documented in this thesis can serve as a reference for similar Flutter-based projects.

All tasks set out in the Introduction were accomplished (Tasks 1-4 correspond to Chapters 1-4). The research hypothesis was validated, and the aim of the thesis has been reached.

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Appendix A. System Code Excerpts

A.1. House Model (Excerpt)

The following code excerpt shows the core House model class with its primary fields and the factory constructor for Firestore deserialization:

```
class House {
  final String id;
  final String title;
  final String? titleRu;
  final String? titleUz;
  final String description;
  final double price;
  final String address;
  final double? latitude;
  final double? longitude;
  final String district;
  final int bedrooms;
  final int bathrooms;
  final double area;
  final bool isRent;
  final bool isVerified;
  final bool isSuspicious;
  final double rating;
  // ... 20+ additional fields

  factory House.fromFirestore(DocumentSnapshot doc) {
    final map = doc.data() as Map<String, dynamic>;
    return House(
      id: doc.id,
      title: map['title'] as String? ?? '',
      price: (map['price'] as num?)?.toDouble() ?? 0.0,
      // ... field mapping
    );
  }
}
```

A.2. NLP Parsing Service (Excerpt)

```
class NLPParsingService {
  static const List<String> _knownLocations = [
    'fergana', "farg'ona", 'фергана',
    'margilan', "marg'ilon", 'маргилан',
    'kokand', "qo'qon", 'коканд',
    // ... 20+ location variants
  ];

  HouseFilter parse(String text) {
    final lowerText = text.toLowerCase();

    // Parse Type (Rent vs Buy) - 3 languages
    bool? isRent;
    if (lowerText.contains('rent') ||
        lowerText.contains('ijara') ||
```

```
        lowerText.contains('аренда')) {
      isRent = true;
    }

    // Parse Rooms: "3 bedroom", "3 xonali", "3 комнатная"
    final roomRegex = RegExp(
      r'(\d+)\s*-?\s*(?:room|bed|xonali|комнат)'
    );
    // ... parsing logic
  }
}
```

A.3. Recommendation Scoring Algorithm (Excerpt)

```
double _calculateScore(House house, HouseRequirements req) {
  double totalScore = 0.0;

  // A. Price Score (Max 40 points)
  if (req.maxPrice != null) {
    if (house.price <= req.maxPrice!) {
      totalScore += 40;
    } else {
      final overPercentage =
        (house.price - req.maxPrice!) / req.maxPrice!;
      if (overPercentage < 0.2) {
        totalScore += 20 * (1 - (overPercentage / 0.2));
      }
    }
  }

  // B. Bedrooms Score (Max 30 points)
  if (req.bedrooms != null) {
    if (house.bedrooms == req.bedrooms) {
      totalScore += 30;
    } else if ((house.bedrooms - req.bedrooms!).abs() == 1) {
      totalScore += 15;
    }
  }

  // C. Location Score (Max 25 points)
  // D. Features & Reputation (Max 15 points)
  // E. Proximity Bonus
  return totalScore;
}
```

Appendix B. Firestore Security Rules

```
rules_version = '2';
service cloud.firestore {
  match /databases/{database}/documents {

    function isAdmin() {
      return request.auth != null &&
        get(/databases/$(database)/documents/users/
          $(request.auth.uid)).data.role == 'admin';
    }

    function isAuth() {
      return request.auth != null;
    }

    function isRealtor() {
      return request.auth != null &&
        get(/databases/$(database)/documents/users/
          $(request.auth.uid)).data.role == 'realtor' &&
        get(/databases/$(database)/documents/users/
          $(request.auth.uid)).data.isVerifiedRealtor == true;
    }

    match /houses/{houseId} {
      allow read: if true;
      allow create, update, delete: if isAuth();

      match /reviews/{reviewId} {
        allow read: if true;
        allow create: if true;
        allow update, delete: if isAdmin();
      }
    }

    match /users/{userId} {
      allow read: if isAuth() &&
        (request.auth.uid == userId || isAdmin());
      allow write: if isAuth() &&
        (request.auth.uid == userId || isAdmin());
    }

    match /leads/{leadId} {
      allow create: if isAuth();
      allow read, update, delete: if isAuth();
    }

    match /reports/{reportId} {
      allow create: if true;
      allow read, update, delete: if isAdmin();
    }
  }
}
```

Apliecinājums / Affirmation

Ar šo es, Sadullaev, apliecinu, ka bakalaura darbs ir izpildīts patstāvīgi, bez citu palīdzības, no svešiem avotiem ņemtie dati un definējumi ir uzrādīti darbā. Šis darbs nekādā veidā nav iesniegts nevienai citai pārbaudījuma komisijai un nekur nav publicēts.

Hereby I, Sadullaev, affirm that the Bachelor's thesis was performed independently; sources of data and definitions are provided. This work has not been submitted to any other examination commission and has not been published elsewhere.

Rīga, 20___. _____. _____

paraksts/signature

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